Class 10 Math - Chapter 7: Coordinate Geometry

🌟 Introduction

Coordinate geometry elegantly bridges the concepts of algebra and geometry. It involves locating points cleanly on a 2D Cartesian plane defined by two perpendicular axes (X-axis and Y-axis).

Every point exists as an ordered pair (x, y), where:

- x is the **abscissa** (horizontal distance from the Y-axis).
- y is the **ordinate** (vertical distance from the X-axis).

📌 Top Formulas In Coordinate Geometry

1. The Distance Formula

To find the direct, graphical distance `d` between any two points $P(x_1, y_1)$ and $Q(x_2, y_2)$: $$ d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} $$

> [!NOTE] **Distance from Origin:** The distance of a point $P(x, y)$ from the central origin $O(0,0)$ is simply $\sqrt{x^2 + y^2}$.

2. The Section Formula

If a point $P(x, y)$ divides the line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the precise ratio $m_1 : m_2$, the coordinates of P are: $$ x =

$$\frac{m_1x_2 + m_2x_1}{m_1 + m_2}, \quad y = \frac{m_1y_2 + m_2y_1}{m_1 + m_2}$$

3. The Midpoint Formula

A special case of the Section Formula where the ratio is 1:1. The midpoint exact center M is: $$x = \frac{x_1 + x_2}{2}, \quad y = \frac{y_1 + y_2}{2}$$

⚡ The Area of a Triangle

If you are given the precise coordinates of the three vertices of a triangle—$A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$—its absolute area is found algebraically: $$\text{Area} = \frac{1}{2} | x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) |$$

> [!IMPORTANT] **Collinearity Condition:** If three points lie on the exact same straight line, the area of the "triangle" they logically form is zero (0). Set the Area formula equal to zero to quickly prove collinearity!

Formula Quick Map

```
graph TD;
    Coord_Geo-->Distance_Formula["Distance Formula\nLengths & Radii"];
    Coord_Geo-->Section_Formula["Section Formula\nRatios & Division"];
    Coord_Geo-->Area_Triangle["Area of Triangle\nSpace enclosed"];
    Section_Formula-->Midpoint["Midpoint Formula\nRatio 1:1"];
```

❌ Common Mistakes

> [!WARNING]
>
> 1. **Subtracting Negatives in Distance:** Applying the distance formula when coordinates are negative. Example: $x_2 - x_1$ when x_1 is -3 becomes $x_2 + 3$. Forgetting this changes everything.

> 2. **Cross-multiplying in Section Formula:** Ensure m_1 multiplies with x_2 (the *opposite* side's coordinate) and m_2 multiplies with x_1. Do not multiply a ratio part by the coordinate sitting right next to it.
> 3. **Modulus in Area:** The area can never be geometrically negative. Even if the calculation within the bracket yields -15, taking the absolute value (15) before multiplying by half is mandatory.

📝 Quick Revision

Shapes in Coordinate Geometry

You can prove the identities of geometric shapes algebraically:

- **Square:** Prove all 4 distances (sides) are perfectly equal AND the 2 diagonals are strictly equal.
- **Rhombus:** Prove all 4 sides are perfectly equal (diagonals will normally be unequal).
- **Rectangle:** Prove opposite sides are strictly equal AND 2 diagonals are completely equal.
- **Parallelogram:** Prove opposite sides are strictly equal (or easily prove their diagonals geometrically bisect each other using the Midpoint formula).

Summary

Coordinate geometry removes the requirement of complex drafting tools and instead relies purely on algebra to define space. Whether computing the shortest path distance, locating the perfect slicing division spot, or verifying a shape, stick strictly to the formula and always draw a quick draft sketch!